Price Discrimination in Over-the-Counter Currency Derivatives*

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Abstract

Through an analysis of transaction level data, this study provides evidence of considerable price discrimination in the Indian over-the-counter (OTC) currency derivatives market. Clients transacting with a single bank (dealer) paid twice as much average markup as that incurred by clients with two dealer counterparties, while the same for clients transacting with ten or more dealers was close to zero. This alludes to the role of bargaining power in pricing, possibly, on account of dealer access. Retail clients (individuals, proprietorship firms and small firms) and unlisted firms were charged a higher markup vis-à-vis listed firms. A majority (83%) of clients transacted with a single dealer counterparty hinting at difficulties related to dealer access. The study makes a case for democratizing market access to enhance competition which may result in better pricing for clients.

JEL Classification: G13, G14, G15
Keywords: Currency derivatives; foreign exchange; OTC markets; price discrimination; bargaining power

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1 Introduction

Cross-border trade in goods, services and assets (or capital) necessitates exchange of one currency into another; and can be thought of as the primary reason why currency markets exist[1]. Based on the asset class traded, currency markets can be classified into spot and derivative segments. Participants trade in spot (including cash and tom) market to meet their immediate requirement of asset (currency), while the derivative contracts are used to hedge against possible adverse fluctuations in exchange rates (currency risk). Currency contracts can be traded at exchanges or over-the-counter (OTC). Exchanges offer superior risk management and anonymity; OTC segment provides choice of counterparty and customization of contract.

Currency market is largely OTC[2]with a daily volume of USD 6.6 trillion[3]globally. Indian onshore currency markets have grown in size and importance as India has increasingly integrated with the global economy[4],with current average daily volume in OTC and exchange traded currency derivative (ETCD) segments at about USD 40 billion and 8 billion respectively[5] For regulatory purposes, currency markets are also categorized as interbank and client (or customer) segments. Banks, which have been granted Authorized Dealer Category I ("dealer") license by the Reserve Bank of India (RBI), are market makers for currency markets and hence require a liquid interbank (or inter-dealer) market for providing quotes to clients and; covering their positions for risk management and compliance with regulatory requirements. The client segment witnesses participation from a diverse investor set; from

[1]Report of Reserve Bank of India (RBI) taskforce on Offshore Rupee Markets, 2019.

[2]Today only about three per cent of foreign exchange is transacted on exchanges in the form of futures and options contracts [https://www.nasdaq.com/articles/forex-market-overview-2019-06-07].

[3]Bank for International Settlements' (BIS) Triennial Central Bank Survey 2019 on Global foreign exchange market turnover.

[4]For instance, trade to GDP ratio has increased from 24% in 1998 to 40% in 2016 . BIS data reveals that during the same period, onshore currency market daily average turnover have grown from USD 2 billion (1998) to USD 34 billion (2016).

[5]BIS Triennial Central Bank Survey 2019 on Global foreign exchange market turnover and RBI Draft report of Internal Working Group on Comprehensive Review of Market Timings, 2019.

sophisticated foreign institutional investors (FIIs) and large corporates to individuals and small and medium enterprises (SMEs).

Studies have revealed that OTC markets with diverse investor set are prone to discriminatory pricing based on sophistication or activity of investors. Study by Duffie et al. (2005) implies that smaller investors, typically those with fewer search options, receive less favourable prices in OTC markets. In US OTC corporate bond markets, trading costs are lower for active institutions relative to inactive ones [Schultz (2001) and OHara et al. (2018)].

There is, however, a dearth of similar literature for the OTC derivatives market, which could possibly be on account of lack of available data. OTC derivatives and limited transparency on activities in such markets have often been cited among the primary reasons for the 2008 financial crisis [Stulz (2010), Stanton and Wallace (2011)]. The G20 and Financial Stability Board (FSB) have expressed commitments to reform the OTC derivatives market and one of their stated aim is to improve transparency in these markets. Since 2009, they have urged regulators to mandate reporting of OTC derivative contracts to Trade Repositories (TR)[6]. Following which, many jurisdictions, including Europe, United States and India, have framed regulations to implement the same.

In a first, Hau et al. (2019) analyzed European Union (EU) TR data to establish price discrimination against non-financial clients in the foreign exchange (forex or FX) derivatives market and found that dealers charge higher spreads from less sophisticated clients. This study follows a similar empirical approach and analyses a set of 2,55,352 USD/INR currency forward contracts, between 8,173 clients and 70 dealers, during a one-year period (September 1, 2018 to August 31, 2019). Regression results show that, in alignment with the findings of Hau et al. (2019), markup (also spread or transaction cost or rent or margin) in the Indian currency forward markets varies systematically with client sophistication measured

[6]Leaders' statement document for the G 20 Pittsburgh Summit (September 24 25, 2009) mentioned that the OTC derivative contracts should be reported to trade repositories.

empirically using the number of dealer counterparties of a client. Clients transacting with a single dealer counterparty (unsophisticated clients) paid, on an average, a markup of around 18 paise while the same for clients transacting with two dealer counterparties was about 9 paise. Average markup incurred by highly sophisticated clients, with access to ten or more dealers, was about zero. This alludes to the role of bargaining power in pricing, possibly, on account of dealer access. Results were further refined by categorizing the clients into retail (individuals, proprietorship firms and small firms), foreign investors, unlisted and listed firms; and analysing the pricing for each category. The analysis shows that retail clients and unlisted firms incurred an average transaction cost of 19 and 11 paise respectively, while, the same for listed firms and foreign investors was 3 to 4 paise.

A large majority (83%) of clients, almost all (96%) retail clients and 78% of unlisted firms, transacted with a single dealer during the observation period. While more than half of listed companies had access to two or more dealer counterparties. Regression results indicate that, across client categories, increased search option is associated with better pricing with higher benefit accruing to retail clients and unlisted firms. Results of econometric modelling indicate that for the 83% clients, with a single dealer counterparty, increasing access to one more dealer can lead to savings in markup of about Rs. 2.27 billion for total transactions with gross notional of USD 80 billion undertaken during the observation period.

These results indicate that unsophisticated clients are discriminated against and are charged higher rent on account of their lower negotiating and bargaining power because of limited search options. While access to more market makers could reduce markup, a large number of clients (83%) still transacted with a single dealer indicating difficulties related to dealer access; or increasing the search options.

The study has important policy implications. The unfair pricing prevalent in the Indian currency markets is detrimental to the interest of clients and there is a case for democratizing market access. One possible way for achieving this is by promoting the usage of electronic dealing platforms to bridge the gap between clients and dealers, improve transparency and enhance competition for better pricing for clients.

The rest of the study is structured as follows: A brief summary of related literature has been provided in the following section. Section 3 describes the process of transacting in the currency forward market. Section 4 presents the hypothesis of the study. Section 5 discusses the data while the description of the variables is provided in Section 6. Section 7 presents the empirical analysis followed by robustness checks in Section 8. Section 9 concludes the study.

2 Related Literature

This study is closely related to the strand of theoretical and empirical literature that seeks to understand the impact on asset prices of search, liquidity, bargaining and informational frictions in decentralized over the counter markets. Theoretically, the role of search costs is analysed in detail in Duffie et al. (2005) who find that bid-ask spreads are set in consideration of investors' outside options which are reflective of accessibility of other market makers and investors' ability to find counterparties. This results in spreads being tighter for more sophisticated investors who have better access to other investors or market makers with limited bargaining power. Duffie et al. (2007) analyse the implication of search frictions for risky asset pricing. Golosov et al. (2014),Bolton et al. (2016), Babus and Hu (2017), Babus and Kondor (2018) provide theoretical evidence on the role of information acquisition and diffusion in determination of asset prices in OTC markets.

Existing empirical studies on price dispersion in over the counter markets have mainly focused on bond markets mainly because of availability of reliable granular data for corporate and, more recently, municipal bond markets. Schultz (2001)) finds that corporate bond trading costs are a decreasing function of trade, investor and dealer size. Harris and Piwowar (2006) and Green et al. (2007) find similar size effects in municipal bond trading market. More recent studies [Hendershott et al. (2020)]; OHara et al. (2018)] have found evidence of price discrimination by analysing trading activity of insurance companies in corporate bond markets where larger and more active insurance companies pay lower spreads. Friewald and Nagler (2019) provide evidence that systematic OTC frictions like market-wide inventory, search, and bargaining frictions account for one-third of the total explained variation in US corporate bond yields. The current study predominantly relates to this set of empirical studies that highlights the role of search and bargaining frictions on asset prices in OTC markets. There is also, however, a significant literature on role of information disclosure and market transparency on trading costs in OTC markets and as such, would provide useful directions for future research in asset pricing in Indian OTC currency derivatives markets. Among the empirical studies in this area are Bessembinder et al. (2006)), Goldstein et al. (2007) and Edwards et al. (2007)) who find that the Introduction of Trade Reporting and Compliance Engine (TRACE) in 2002 for US corporate bond markets that facilitated mandatory reporting of OTC secondary market transactions, improved bond market liquidity and reduced trading costs. Similar effects of transparency on bond market trading costs for US municipal bonds were found by Schultz (2012). Credit Default Swap (CDS) markets also experience lower transactions costs and improved liquidity following enhanced public dissemination of trading information [Loon and Zhong (2016) and Bhat et al. (2016)].

The current study contributes directly to the scant but growing literature on studying price dispersion in derivatives market. Using unique European data on interest rate swaps market, Cenedese et al. (2020)) find substantial and persistent heterogeneity in derivative prices. They find that OTC premia are lower for contracts cleared via central counterparty, for clients posting initial margins and clients which have high creditworthiness. Premia are absent for dealers which is suggestive of dealers possessing significant bargaining power in these markets. As mentioned earlier in the text, the current study closely relates to Hau et al. (2019) that provides evidence of significant price discrimination against unsophisticated investors in European currency derivatives market. To our knowledge, this is the first study that provides evidence of price discrimination in Indian OTC markets, in general, and currency derivatives market, in particular.

This paper also contributes to the recent literature on intermediary asset pricing [e.g. He and Krishnamurthy (2013), Adrian et al. (2014), He et al. (2017)] that provides evidence of the crucial role that financial intermediaries play in the pricing of financial assets. In particular, the current study empirically highlights the role of OTC derivative dealers in creating heterogeneity in derivatives prices.

A significant result in the paper highlights the difference in mark-ups charged from institutional vis-a-vis retail clients. This finding is related to a larger literature, predominantly tested empirically in equity markets, that highlights the difference in brokerage commissions charged from different categories of investors. Edmister and Subramanian (1982) provide some insights into the determinants of brokerage commission rates in US equity market and Goldstein et al. (2009) demonstrate some size effects with smaller institutions concentrating their trading (with a smaller set of brokers) more than larger institutions and paying higher per-share commissions.

At a macro level, this study is also related to the literature on currency derivative markets in India. Research in Indian derivative markets has been focused on understanding the determinants of usage of currency derivatives [Raghu and Shanmugam (2014) and PJ et al. (2017)] and impact of currency derivatives on exchange rate volatility [Kumar (2015) and Singh and Tripathi (2016)]. Understanding the pricing of these contracts based on search, liquidity, and bargaining frictions remains to be explored in the Indian market. It is hoped that this study would spur future research in this direction in India and other emerging markets.

3 Contract booking in currency forward market

Prices for USD/INR forward contracts are quoted in the market in *Rupees per USD*. A client can request for quote (one-way) for booking a currency forward contract by contacting a dealer directly, for instance, over the phone or by visiting a branch. Quotes can also be requested through certain proprietary electronic dealing platforms of individual banks and Multi-Bank Portals (MBPs); however, access to them is restricted to clients with a minimum order size.

Generally, a dealer provides quote by adding a certain markup over the prevailing interbank rate for a similar tenor contract. Interbank rate is taken as the reference rate since it is the rate at which the dealer covers the currency risk, which it takes over from the client. Markup is the spread or margin which is charged by the dealer to compensate for a number of considerations, which might include risks taken, costs incurred, and services rendered to a particular client.[7] The calculation methodology for markup is not uniform and varies from dealer-to-dealer. Nevertheless, the markup is expected to take into account certain components like the cost for processing documents, counterparty risks, providing advisory services etc. The relative difficulty of hedging a forward transaction in the interbank market, depending on the liquidity of the contact's tenor, is also possibly passed on to the clients through markup.

[7]FX Global Code Principle 14.

4 Hypothesis

As mentioned earlier, the Indian currency derivatives market witnesses participation from a diverse investor set (from SMEs to large corporates) and thus, based on existing theoretical and empirical literature on OTC markets, is prone to discriminatory pricing based on sophistication or activity of investors. Accordingly, the hypothesis adopted for this study is as follows:

Hypothesis: *The markup, in Indian currency forward market, depends on sophistication of clients, with less sophisticated client incurring higher markup for booking currency contracts.*

5 Data

5.1 Description

Clearing Corporation of India Limited (CCIL) operationalized TR services for reporting of OTC currency derivative transactions in 2012[8] . As per RBI guidelines, all dealers are required to report currency derivative transactions to CCIL TR.

Transaction level data for OTC currency derivative transactions executed during a one-year period from September 1, 2018 to August 31, 2019, retrieved from CCIL TR, was used for this analysis. Transactions which were active during the month of September, 2019 or had been settled till then were retrieved. Refintiv Eikons (Eikon) USD/INR spot and interdealer outright forward prices were obtained for calculation of markup. The choice of sample period was guided mainly by two reasons. A one-year period provided sufficient number of observations to draw meaningful conclusions for the Indian currency forward market.

[8] See CCIL Trade Repository (TR) Services [https://www.ccilindia.com/Pages/TradeRepository/TR.aspx]

The dataset was contemporary and any results/conclusions drawn from it were likely to be representative of the current microstructure of the market. Additionally, the data series used in the paper from Eikon was only available for this particular time-period.

Movement in daily closing prices of USD/INR spot and outright forwards for 1 month (1M), 3 months (3M) and 1 year (1Y) tenors during the sample period is provided in Figure 1.

The transaction dataset comprised outright forward transactions. Each transaction level dataset provided information related to - both counterparties (through name); trade date and time and; currency pair, exchange rate, settlement date and notional value of the contract. The analysis was restricted to USD/INR deals, which is the most liquid pair in onshore market.

While transactions for individual person (including joint account or proprietary firm) are permitted beyond interbank market hours[9], interbank rates for such time periods are not available for markup calculation. Hence, these transactions were filtered out. Onshore currency forward market is fairly illiquid for tenors exceeding one year[10], therefore, transactions with tenor exceeding 365 days were omitted from this analysis. For the remaining, a truncation was done to remove trades with outlier spreads (falling in the top and bottom 0.1 percentile) and notional amount (top 0.1 percentile).Table 1 provides the detailed results of step by step filtering applied on the data. Finally, a set of 2,55,352 trades, between 8,173 clients and 70 dealers, was analysed for this study.

5.2 Summary statistics

Table 2 reports summary statistics. The first panel shows information pertaining to the 8,173 clients who have transacted during the observation period. The client at 90th percentile is

[9] Foreign Exchange Dealers' Association of India (FEDAI) Rules.
[10] CCIL Rakshitra.

charged about thirty five times the markup incurred by the client at 10th percentile, hinting at possible price discrimination. With regard to dealer counterparties, more than 75% of the clients have transacted with a single dealer and even the 90th percentile client has entered into transactions with only two dealers, implying possible frictions in accessing market makers for majority of the clients. The clients in top 0.1% bracket have access to 15 to 29 dealers. The gross notional amount traded for the client at 10th and 90th percentile is about 0.1 million and 26 million respectively. The mean gross notional of 49.2 million is higher than that of 90% of the clients indicating a highly skewed distribution. Further, the median client has booked a total of 6 contracts while client at 90th percentile has booked 63 contracts, out of the total 2,55,352 transactions analysed.

The bottom panel (transaction data), in Table 2, describes data on 2,55,352 trades executed during the observation period. The average spread over all trades is 9.5 paise, lower than the average spread for clients of 21.2 paise. Further, more than half of the transactions during the analysis period were undertaken by clients having 2 or more dealer counterparties, who account for less than a quarter of the total clients, implying the presence of a small set of active clients. The median transaction has a tenor of 56 days. A frequency distribution of tenor of the contracts is provided in Figure 2. The average notional value of a contract is USD 1.6 million with a standard deviation of 5.2 indicating a significantly high dispersion in requirements. Additionally, the data also indicates that the client was long USD in majority (56%) of the contracts.

6 Variables

6.1 Markup

The *markup* (in paise) for a transaction, τ, has been calculated using the following formula:

$$markup_\tau = B_\tau * (F_\tau - I_\tau) * 100 \tag{1}$$

where B_τ is 1 or -1 depending upon whether the client is long or short USD respectively. F_τ is the exchange rate of the forward contract, as specified in the transaction data set. I_τ is the estimated contemporaneous interbank outright forward mid-price.

For calculating I_τ, high (H) and low (L) prices, for spot and outright forward contracts, were retrieved for standard half-hourly intervals (9:00 to 9:30 am, 9:30 to 10:00 am, 10:00 to 10:30 am and so on) from Eikon. This set of data is available for standard tenors (overnight, 1 month, 2 months and so on). For a transaction (τ) executed on date d and at time t, high (H_t) and low (L_t) prices for a similar tenor interbank outright forward contract prevailing on d during the standard half-hour, around t, were considered. I_τ was then calculated as,

$$I_\tau = (H_t + L_t)/2 \tag{2}$$

In cases of non-standard tenors, interpolation from nearest standard tenors was used for arriving at H_t and L_t.

6.2 Client sophistication

During the one-year observation period, the number of dealer counterparties with which a client has traded in USD/INR forwards, #Dealers, has been considered as the measure of client sophistication. This is based on the inference by Duffie et al. (2005) that smaller investors, typically those with fewer search options, receive less favourable prices. #Dealers has been used to quantify the search options of a client, with a more sophisticated client having better access to market makers (thus, higher value of #Dealers) and, thereby, commanding more bargaining power relative to a less sophisticated client, who has fewer search alternatives (or lower value of #Dealers). Thus, client sophistication rises with increase in dealer counterparties (#Dealers).

6.3 Client classification/categorisation

Investors/clients in financial markets are generally classified as retail and institutional clients. While these are amorphous terms which are frequently used with varying meanings, in this study, an attempt has been made to categorize clients, into retail and institutional, based on identities.

The clients in the data were categorized into following functional categories 1. Retail Clients; 2. Unlisted companies; 3 Listed companies; 4. Foreign Investors and; 5 Related entities (offshore branches/offices or related entities of dealer banks). Retail clients category consists of individuals, sole proprietorships, partnerships and small companies which were not covered in CMIE Prowess database[11].The classification of clients into listed and unlisted

[11]Prowess is a comprehensive database that covers over 50,000 Indian companies. It includes the universe of all companies traded on the National Stock Exchange and the Bombay Stock Exchange, a large number of unlisted public limited companies and private limited companies. The data in Prowess is derived from Annual Reports, quarterly financial statements, news feeds from exchanges and other varied sources and doesnt suffer from a deliberate survivorship bias. Most academic studies on corporate finance in India use CMIE Prowess as their main source for data. We include the companies/LLPs/Partnerships that did not have a CIN and couldn't be matched to Prowess data in the retail category as these are likely to be smaller

categories is based on CIN numbers obtained by matching the client name from CCIL data with that in CMIE. Foreign investors category mainly has FIIs, but also few clients with direct investments. The data for FIIs was taken from Central Depository Services Limited (CDSL). There were five clients who were part of Prowess database but had no reported CIN (classified as Others). The number of clients and their share in total traded notional in reported in Table 3.

6.4 Contract parameters

A dealer has to follow certain standard operational procedures, like processing client documentation etc., while booking a forward contract. The fixed cost, incurred by a dealer, on account of such procedures is possibly passed on to the client through markup. This fixed cost could be distributed in markup as,

fixed cost (in Rupee) / notional value (in USD) of the contract.

A dealer covers the currency risk, which it takes over from the client, through an opposite position in the interbank market. Therefore, transaction with notional value less than interbank lot size exposes the dealer to market risk, as it cannot be immediately covered in interbank market. Dealers generally collate several small transactions and cover the risk once the gross notional of such transactions become dealable in the interbank market. Hence, Notional value of the contract is expected to inversely influence markups. Further, contracts with settlement date coinciding with month-end, which are standard tenors and fairly liquid, are relatively easier to cover. Therefore, Customization of contract, i.e. the difference in days of the settlement date, of the forward contract, from the nearest month end, could increase the markup. Liquidity in the forward market dries up as the tenor increases, hence, the markup may increase with the Tenor of the contract. Volatility in exchange rate will

companies that do not have mandatory disclosure requirements.

also impact dealer's ability to cover his position. Hence, Rogers-Satchell Volatility, based on extreme values (open-high-low-close of spot prices) during each half hour interval has been used. The volatility numbers thus obtained are multiplied by 10^4 while estimating the regression models in order to normalize the coefficients. A long or short position in USD may not be equivalent for a dealer on account of factors like asset liability management, capital charges etc. Hence, a Sell dummy variable, with values 1 and 0 depending upon whether the client is short or long USD respectively, has been used to capture the impact of buying or selling USD on spread.

6.5 Dealer heterogeneity

As has been mentioned, markup calculation methodology is not uniform across dealers. Despite accounting for client sophistication and contract characteristics, there are bound to be certain dealer-specific characteristics which influence markup. Hence, dealer fixed effect, to account for dealer heterogeneity, has been used in the model.

6.6 Date and time fixed effects

Date fixed effect has been used to capture the impact of date-specific effects. Research, for instance Khademalomoom and Narayan (2019), exhibit that time-of-day, because of factors like overlap with other major markets, influence the behaviour of currency returns. In the model, this has been accounted for through minute of day fixed effect.

7 Analysis

7.1 Client Sophistication

Figure 3 plots average markup paid by clients trading with a specific number of dealer counterparties. It indicates that as client sophistication increases, the markup decreases. Clients transacting with a single dealer counterparty (unsophisticated clients) paid, on an average, a markup of around 18 paise while the same for clients transacting with two dealer counterparties was about 9 paise. Average markup incurred by highly sophisticated clients, with access to ten or more dealers, was about zero.

It is noteworthy to mention that more than 83% of the clients have transacted with only one dealer counterparty during the study period, and are therefore, relatively unsophisticated.

7.1.1 Model

A linear model, similar to the one developed by Hau et al. (2019), albeit with certain modifications was used for estimating markup. The specification is as provided below

$$markup_{\tau,i,d,t,m} = \beta_1 S_i + \beta_2 C_\tau + \gamma_d + \lambda_t + \Omega_m + \epsilon_\tau \tag{3}$$

where $markup_{\tau,i,d,t,m}$ is the markup for transaction τ between client i and dealer d on date t at time (minute of the day) m. S_i is the measure of client sophistication (as defined in section 6.2), C_τ is a vector representing contract specification and takes the form *[LogTenor, LogNotional LogCustomization, Volatility, Sell]*. Dealer-specific (γ_d) date-specific (λ_t) and minute-of-day (Ω_m) fixed effects have also been used.

7.1.2 Results

The regression results (Table 4) substantiate the hypothesis that dealers charge higher markup from less sophisticated clients while booking currency forward contracts. It can be observed from the results that the value of markup is negatively related to the degree of client sophistication, or the number of dealer counterparties of a particular client, and that this relation is statistically significant at 1% level of significance. This effect was found consistently across all regression specifications.

With regard to contract specifications, as expected, longer tenor and higher degree of customization increase the markup. It was also observed that higher markups were charged for transactions where the client was short USD indicating that dealers charge a premium when clients are selling dollars and buying rupees in the forward market. Results also indicate that markups decrease with increasing notional value of contract, in accordance with the practice of fixed cost charged per transaction. Volatility does not seem to have a significant impact on markup. It's magnitude goes down after inclusion of date and minute-of-day fixed effects. This could be because variation in daily volatility gets absorbed in these fixed effects.

7.2 Client Categorisation

It can be observed from Figure 4, which shows average markup charged for different client categories, that a wide dispersion exists in the transaction cost incurred by clients. While average spread for retail clients remained high at about 19 paise, the same for related entities, foreign investors and listed entities was in the range of 3 to 4 paise. Average margin for transactions involving unlisted firms was 11 paise. Further, as the proportion of clients with access to two or more dealers increases for a category, the average spread decreases, in alignment with the findings in the previous section (7.1). However, foreign investors and related entities appear to be an exception. Because of low number of clients in the others category, it has been excluded from this section.

A deeper look at the category of foreign investors and related entities was taken to understand if there were some other unobservable constraints that were guiding their use of limited dealer counterparties. It was found, that, on account of regulatory and business-related factors, an FII invests, across multiple asset classes, in the Indian markets through arrangements with an onshore custodian bank. It was observed that all custodian banks are also dealers, and generally FIIs booked USD/INR currency forwards with their respective custodian banks[12]. Therefore, while FIIs are fairly sophisticated investors, their arrangement with a single dealer is on account of factors not related to lower search and bargaining power. Within foreign investors category, clients transacting with multiple dealers were those with direct investments. The same is also true for related entities, wherein the offshore branches/related entity of a dealer transacted with their dealer counterparts.

7.2.1 Model

To control for various contract-specific, dealer-specific or date/time-specific factors, the following linear model was used

$$markup_{\tau,i,d,t,m} = \zeta_i + \beta_2 C_\tau + \gamma_d + \lambda_t + \Omega_m + \epsilon_\tau \tag{4}$$

where $markup_{\tau,i,d,t,m}$ is the markup for transaction τ between client i and dealer d on date t at time (minute of the day) m. ζ_i is the dummy variable depicting client category, C_τ is a vector representing contract specification and takes the form *[LogTenor, LogNotional LogCustomization, Volatility, Sell]*. Dealer-specific (γ_d), date-specific (λ_t), and minute-of-day (Ω_m) fixed effects have also been used.

[12]During the study period.

7.2.2 Results

Results from regressions based on client category dummies are reported in Table 5. Related entities is the omitted client category dummy for these regression specifications as they are expected to receive the market price for the currency pair, or a price very close to market price. The results were consistent with this hypothesis. From the results, it was observed that retail clients were charged the highest markup since they were likely to be the least sophisticated investors in this market given their limited resources, networks and expertise (and hence, bargaining power) in trading. The next highest markup was charged from unlisted firms followed by foreign investors. Listed entities did not receive a markup that is significantly different from that received by related entities indicating high sophistication.

In Table 6, separate category wise regressions, to estimate the impact on markup caused by an increase in bargaining power through access to a larger number of dealers for each of these client categories, was run . Results were found to be consistent with the hypothesis. For each client category, an increase in the number of dealers reduces markup charged significantly. As expected, retail clients gain the most by having access to a larger number of dealers. Similar effects of improved dealer access on markups are found for unlisted firms and listed firms.

7.3 Economic implications

The 83% of clients, who have transacted with a single dealer counterparty, and contributed about one-fifth to the gross notional have incurred more than half of the total transaction costs (Table 7). Further, retail clients and unlisted firms contributed to less than one-third of gross notional value of transactions but their share in total markup was about two-third

(Table 8).

A majority (83%) of clients have transacted with a single dealer counterparty. To understand the price improvement on increasing access to one more dealer for clients, transacting with a single dealer in each category, the following calculation was made,

$$PriceImprovement = log2 * (\beta_i * GrossNotional_i), \tag{5}$$

where $beta_i$ is the coefficient estimated for the effect of client sophistication on markup for respective client category (Table 6). $GrossNotional_i$ is gross notional amount (in USD mn) for all transactions by the specific client set. The calculation indicates that increasing access to two dealer counterparties can lead to price improvement by half for retail clients, one-third for unlisted firms and one-fifth for listed firms. In general, for the 83% unsophisticated clients, increasing access by one dealer can lead to price improvements or savings in markup of about Rs. 2.27 billion on total transactions with gross notional of USD 80 billion undertaken during this period (β_i from Table 4).

8 Robustness Tests

A series of robustness tests were conducted to check the consistency of the main results in the paper and ensure that the results are not sensitive to small changes in sample period, client category, variable definition and control variables specification.

8.1 High Vs. Low Volatility Periods

The observation period was segregated into cycles of high and low volatility based on daily rolling three months realised volatility data (from Eikon) which measures the actual dispersion of returns of the currency pair over the last three months based on daily high and low prices. For the observation period considered in this study, the three months interval of October 10, 2018 to January 9, 2019 (Figure 5) was a period of high volatility and volatility was the least during the interval of May 2, 2019 to August 1, 2019. During the two volatility cycles, results were found to be consistent with what was obtained earlier (Table 9).

8.2 Financial Clients

There were 87 financial clients (including FIIs and related entities) in the sample who engaged in a total of 4346 transactions during the sample period. They constitute a small percentage (1.06%) of total clients and (1.7%) total transactions in the sample. Nevertheless, in order to test whether the impact of client sophistication on markup is being driven by trades of financial entities, regressions were rerun by excluding all financial clients. The results and analysis are reported in Table 10. It is observed that the results remain quantitatively and qualitatively unaffected by this exclusion.

8.3 Additional Measures of Sophistication

Additional measures of sophistication were used to test whether results were sensitive to the definition of client sophistication. In addition to the number of dealer counterparties (Log#Dealers), two more measures were used - total number of transactions entered into by a client (Log#Transactions) and total gross notional of all forward contracts traded by a client (LogGrossNotional) during the sample period. The larger the number of transactions, the more sophisticated a client is likely to be because of informational advantage on account of trading experience. Also, the higher is the trading volume of a client, the higher is the gain from negotiating with dealers to receive lower markups. A sophistication index based on the first principal component of Log#Dealers, Log#Transactions and LogGrossNotional was also constructed. The results and analysis for these three additional measures of client sophistication are presented in Table 11. As can be observed, the results are qualitatively similar to the ones obtained using number of dealer counterparties as a measure of sophistication. Although the results in the paper are robust to alternative definitions of client sophistication, number of dealers has been used as a primary measure because, unlike total number of transactions and gross notional that are governed by investors' business requirements, access to dealers is an external factor that is a choice variable for the clients and can also serve as a target variable that can be influenced by policy makers.

8.4 Dealer Client Relationships

The impact of dealer client relationships on spread was also analysed. It is possible that the set of clients in the current sample source multiple business activities to the dealer banks and hence, interact with the dealers in multiple financial markets in addition to currency markets. This can result in cross-subsidization and may affect the markup paid by clients in currency markets. Existing literature in relationship trading in OTC markets suggests that dealers provide a discount to clients with whom they have a non-exclusive relationship [Hendershott et al. (2020)]. In order to test for the impact of relationship trading, the dealer client interaction in bank credit market was analysed. The dealer client pair in the current sample were matched to Prowess firm level data which provides information on whether a client/firm had listed a dealer bank as their banker in external credit market. Relationship data on 948 clients involving 82,412 transactions was available. This data is biased in favour of sophisticated clients because Prowess does not have information on small retail clients. In this matched sample, average number of dealers that each client trades with is 2.3 which is larger than the average for the full sample (1.4).

The spread was regressed on a relationship dummy which takes a value of one if the dealer was listed as a banker in Prowess data for a given client and zero, otherwise. Contract characteristics, dealer, date and minute of day fixed effects as regressors were also included. The regression results are reported in Table 12.

The regression results in column 1 show that relationship dummy is negative but statistically insignificant. Introduction of Log#Dealers as a measure of client sophistication in regression specification results in relationship dummy being significant at 5% level of significance (Column 2) hinting at discount associated with relationship trading. The discount could also be because this sample focuses mostly on sophisticated clients who have access to larger number of dealers and hence, are unlikely to pay a larger markup associated with being a captive client of a dealer. Introduction of an additional interaction term between relationship dummy and Log#dealers renders the relationship coefficients insignificant (Column 3). The coefficient of the measure of sophistication used in this study, though, remains negative and significant in the presence of relationship dummy hinting at benefits of better access to dealers even after accounting for price impact of relationship trading.

9 Conclusion

Through an empirical analysis of USD/INR forward transactions, the study brings to light the discriminatory pricing, detrimental to the interest of small clients, prevalent in Indian OTC currency derivatives market with higher markup being charged from less sophisticated clients.

Retail clients (individuals, proprietorship firms and small firms) and unlisted firms contributed to less than one-third of gross notional value of transactions but their share in total markup was about two-third. Thus, they are more disadvantaged vis-à-vis other client categories in this market.

Increased access to dealers improves negotiating power of clients and, as per results in this study, improves transaction cost. Despite this, a large number of clients (83%) transacted with a single dealer hinting at difficulties related to dealer access.

Recent studies have highlighted that the use of multi-dealer platforms eliminates price discrimination by increasing dealer competition, with the largest benefits accruing to the least sophisticated clients. Considering that similar outcome is desired in the Indian markets, there is a case for democratizing market access through increased usage of electronic trading platforms, as envisioned by G20 leaders in their statement at the Pittsburgh Summit.

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Figure 1: USD/INR Spot and Outright forwards

The chart shows movement in daily closing price of USD/INR spot and outright forwards for 1 month (1M), 3 months (3M) and 1 year (1Y) tenors during the observation period of this study. Rupee traded with depreciating trend during the initial days (and H1 of 2018-19), on account of rising crude oil prices and tightening of global financial conditions due to increase in Federal funds rate, and touched a low of 74.4 per USD during October, 2018. The Rupee thereafter recovered in Q4 of 2018-19 and remained rangebound during Q1 of 2019-20. During July, 2019 Rupee remained in the range of 68-69 per USD but depreciated to 72 levels in August, 2019, with geopolitical tensions and risk-off sentiments. The average daily closing 1 month, 3 months and 1 year forward points during the period were 26.93, 76.64 and 296.10 paise respectively. (Source: Eikon)

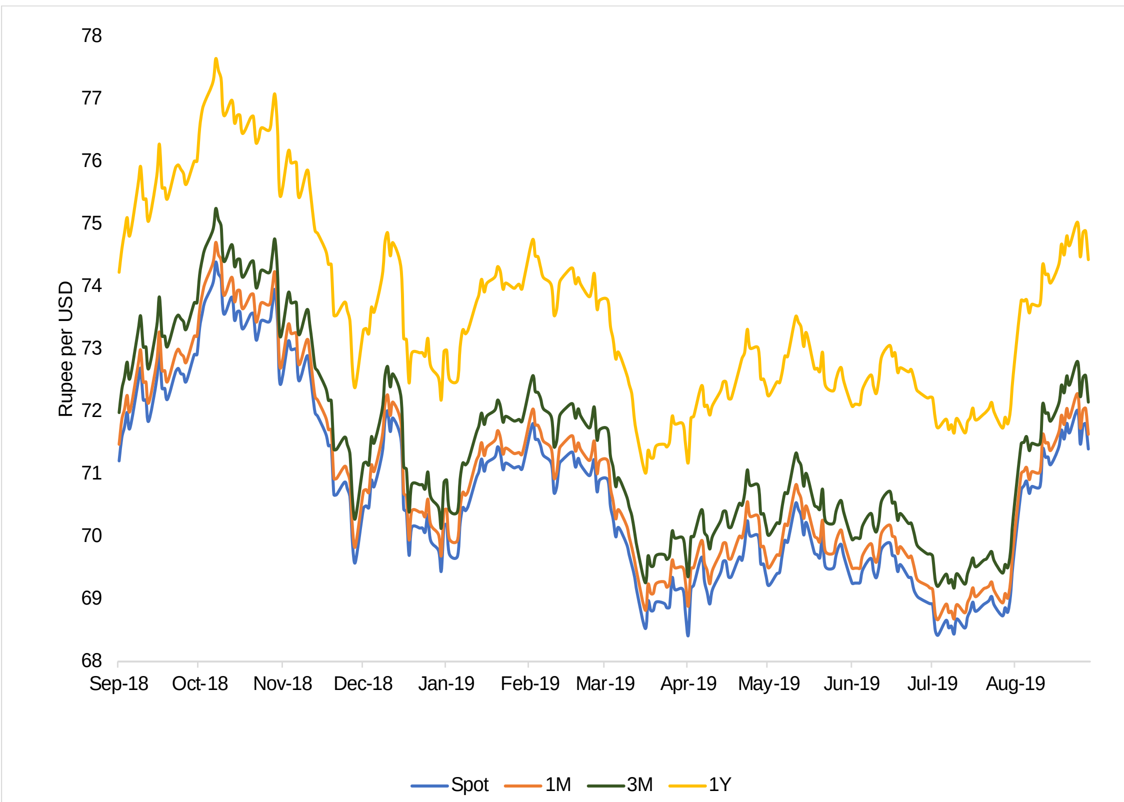

Figure 2: Frequency Distribution of Transactions by Tenor

The chart shows frequency distribution for USD/INR forward transactions based on tenor for transactions with settlement date between spot and 1 year.The horizontal axis shows the number of days in tenor and vertical axis indicates the count of transactions for each tenor.

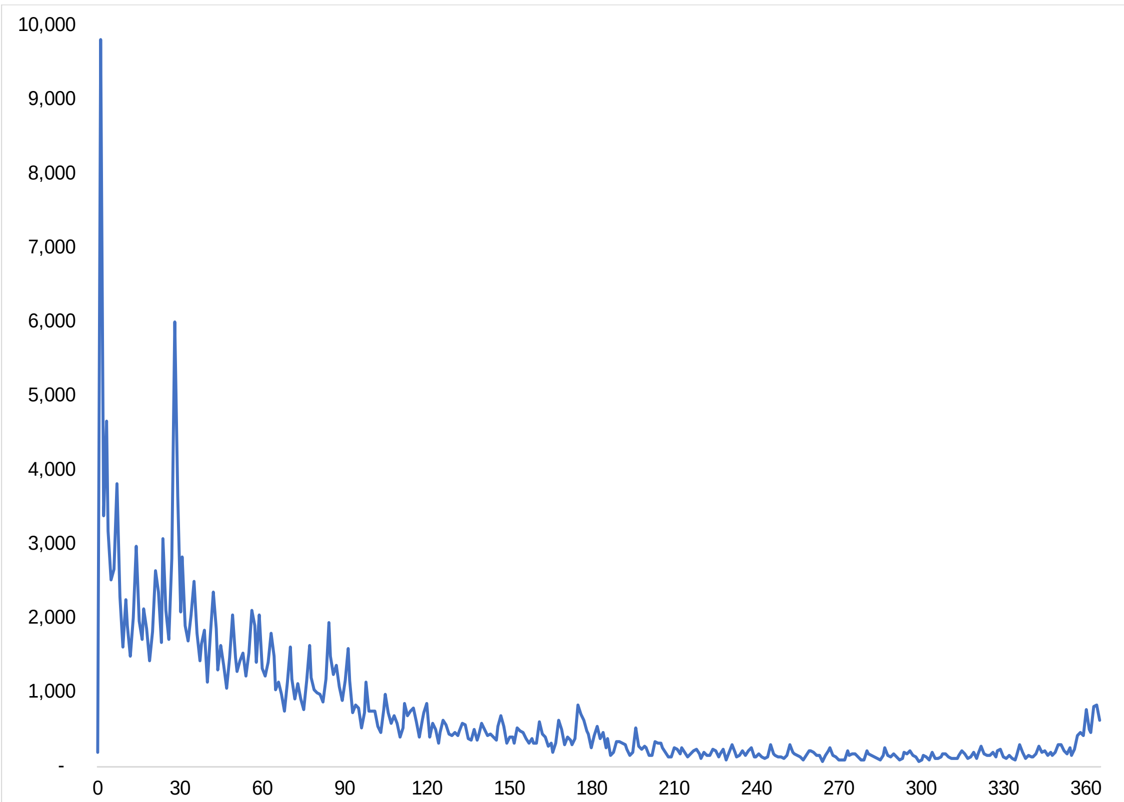

Figure 3: Average Markup vs Number of Dealer Counterparties

The figure plots the average markup paid by clients trading with a specific number of dealer counterparties for USD/INR forward transactions. 10 or more dealer counterparties has been depicted through the number 10 on the x-axis. Size of bubble represents relative value of gross notional traded. Labels indicate the percentage of clients with a given number of dealer counterparties.

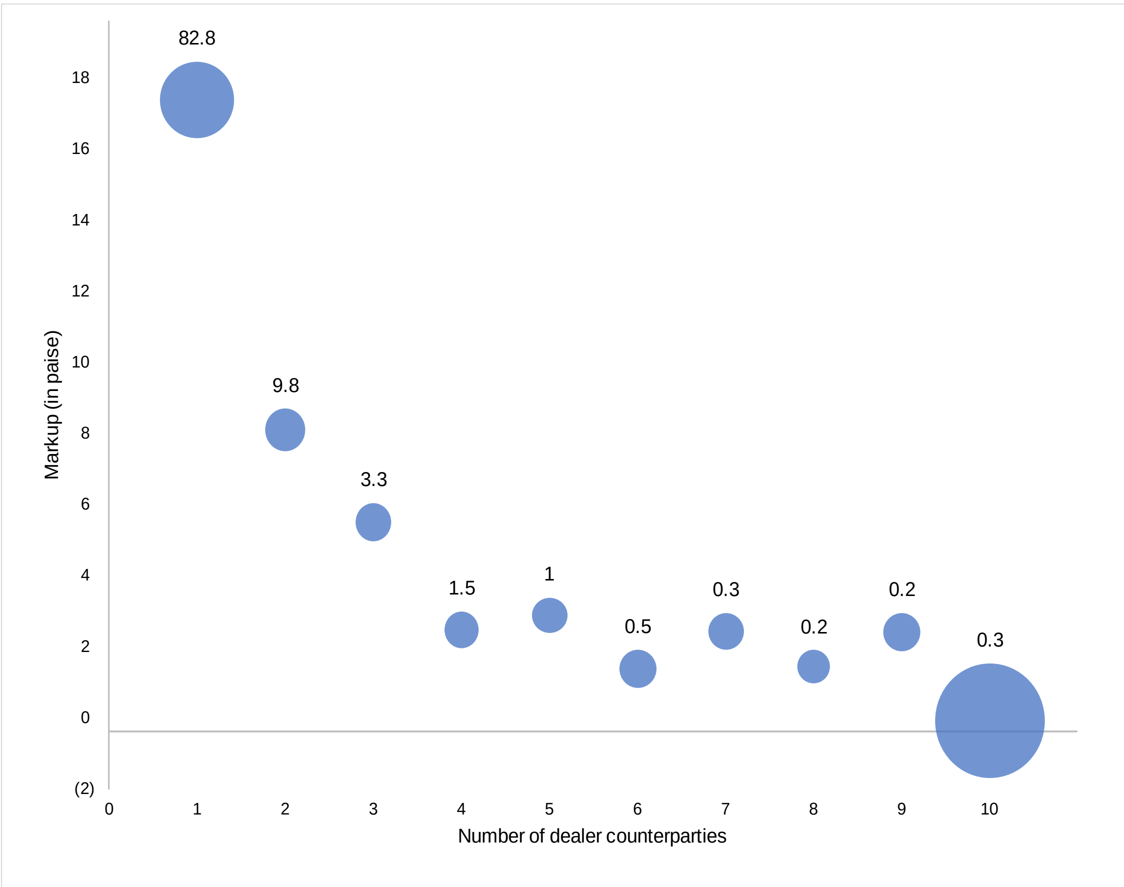

Figure 4: Average Markup vs Percentage of Clients with Two or more Dealer Counterparties

The figure shows a plot of category wise average markup charged vs. the percentage of clients transacting with two or more dealer counterparties for each client category. Data labels mention the average markup.

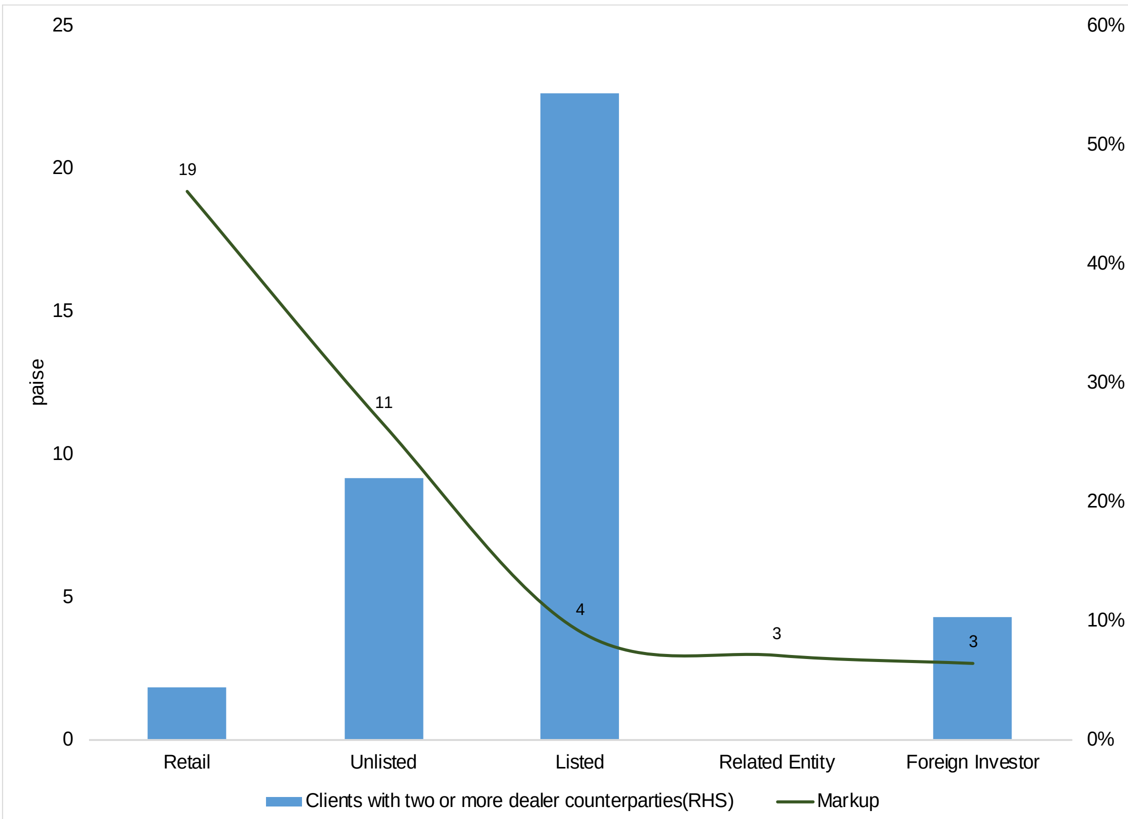

Figure 5: Three Months Realised Volatility

The figure plots the movement of daily rolling three months realised volatility from January 1, 2015 to October 31, 2019 for USD/INR spot, obtained from Eikon. It measures the actual dispersion of returns of the currency pair over the last three months based on daily high and low prices.

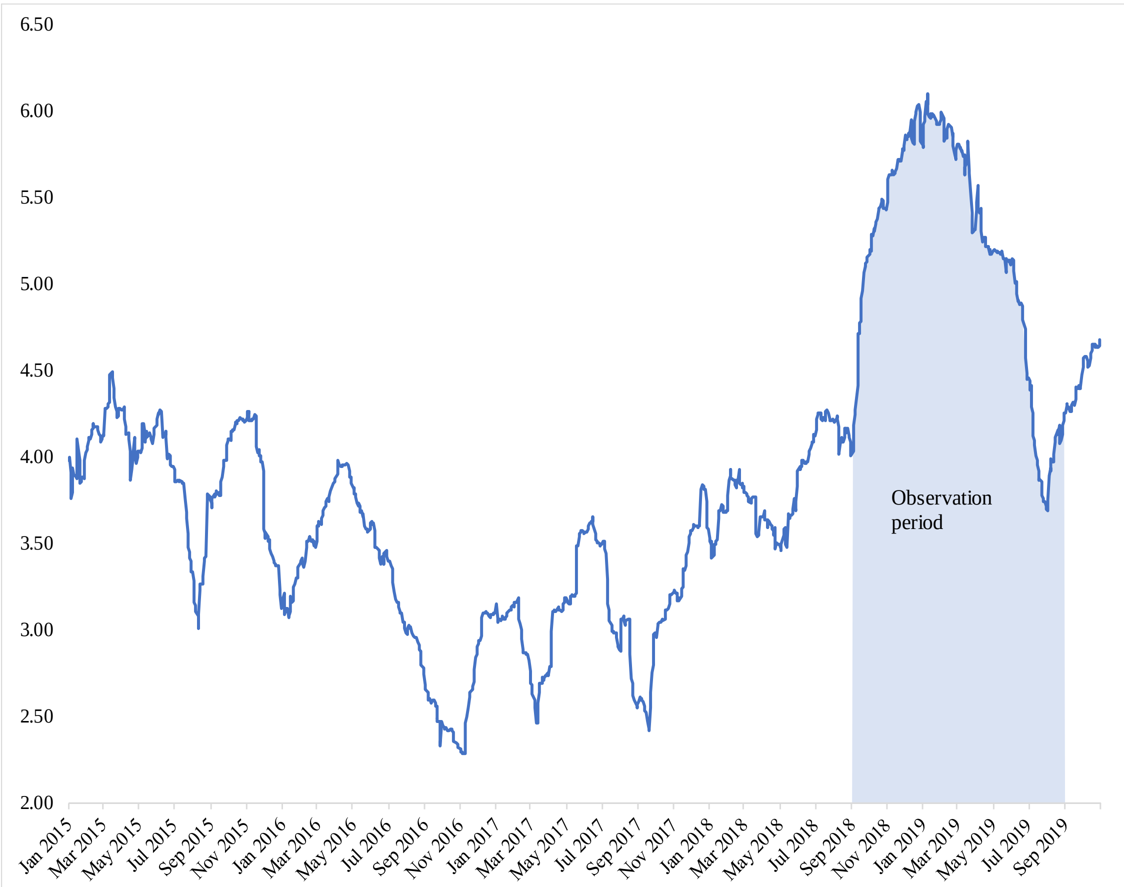

Table 1: Results of Filtering

The table shows the number of observations dropped as per the filtering steps mentioned above. Forward time option deals where the client has the option of settling forward transactions during a specific period instead of a specific date have been dropped in the first step to consider only plain vanilla forward.

Total USD-INR forward deals	3,77,379
(-) Forward time option deals	76,538
(-) Trades with timestamp beyond interbank market time	30,051
(-) Trades exceeding 1 year maturity	14,668
(-) Trades with outlier spreads	770
Total considered	2,55,352

Table 2: Descriptive Statistics

The panel on client data shows data related to the 8,173 clients which have entered into at least one USD/INR forward transaction during the observation period. Average client markup is the average markup that a client pays on its transactions. Dealer counterparties indicate the total dealers with which a client has entered into forward transactions. Gross notional and number of transactions indicate the total notional value and number of contracts respectively booked by the client.

The panel on transaction data shows key statistics for the 2,55,352 transactions examined in the study. Markup is the markup for a transaction. #Dealers indicate the total number of dealers with which a client has transacted during the observation period. Tenor and notional imply the tenor and notional value respectively for a transaction. Customization is the difference of settlement date of the transaction from the nearest month end date. Rogers-Satchell volatility has been used to measure volatility. The values of volatility derived as per the formula have been multiplied by 10^4 for consistency of regression coefficients

	Observations	**Mean**	**sd**	**p10**	**p25**	**p50**	**p75**	**p90**
Client data								
Average client markup (paise)	8,173	21.2	24.9	1.4	5.3	13.0	30.9	50.2
Dealer counterparties	8,173	1.4	1.2	1	1	1	1	2
Gross notional (USD mn)	8,173	49.2	973.7	0.1	0.2	0.9	5.3	26.0
Number of transactions	8,173	31.2	149.0	1	2	6	22	63
Transaction data								
Markup (paise)	255,352	9.5	19.5	-5.9[a]	0.2	5.3	14.8	31.9
#Dealers	255,352	4.3	5.6	1	1	2	5	9
Tenor	255,352	88.1	91.0	6	24	56	120	227
Notional (USD mn)	255,352	1.6	5.2	0	0.1	0.2	0.7	3.5
Customization (days)	255,352	5.6	5.1	0	1	4	10	14
Volatility	255,352	0.54	1.45	0.07	0.15	0.30	0.58	1.09
Sell	255,352	0.4	0.5	0	0	0	1	1

[a]Existing empirical studies Hau et al. (2019) also find negative spreads in their data which they mention is plausible due to literature on price dependence of inventory positions of market makers.

Table 3: Client Categorization

The table provides client category-wise distribution of the number of clients and notional amount. Notional represents the relative value of gross notional traded. The total number of clients in each category is mentioned in #Clients.

Client Category	**Notional (share)**	**#Clients**
Listed	57%	742
Unlisted	28%	3,773
Foreign Investor	10%	58
Retail	3%	3,586
Related Entity	2%	9
Others	1%	5
Grand Total	**100%**	**8,173**

Table 4: Model Results (Client Sophistication)

This table describes results for regressions of the markup on measures of client sophistication and contract characteristics. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	(1)	(2)	(3)	(4)
Client specific characteristics				
Log#Dealers	-4.30***	-4.24***	-4.17***	-4.10***
	(0.53)	(0.52)	(0.51)	(0.50)
Contract specific characteristics				
LogTenor	2.45***	2.43***	2.39***	2.37***
	(0.27)	(0.27)	(0.26)	(0.26)
LogNotional	-0.84***	-0.84***	-0.86***	-0.86***
	(0.16)	(0.15)	(0.15)	(0.15)
LogCustomization	1.67***	1.66***	1.64***	1.64***
	(0.24)	(0.23)	(0.23)	(0.23)
Volatility	-3.73	-3.44	-1.34	-0.77
	(4.79)	(4.77)	(4.22)	(4.21)
Sell	7.86***	7.87***	7.53***	7.54***
	(0.54)	(0.53)	(0.54)	(0.53)
Fixed Effects (FEs)				
Dealer FE	Yes	Yes	Yes	Yes
Date FE	No	No	Yes	Yes
Minute-of-day FE	No	Yes	No	Yes
Adjusted R-squared	0.23	0.23	0.24	0.24
Observations	255,352	255,352	255,352	255,352

Table 5: Model Results (Differential Pricing)

This table describes results for regressions of the markup on client category and measures of contract characteristics. Related entities is the omitted client category. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	(1)	(2)	(3)	(4)
Client categories				
Foreign Investor	2.74*	2.69*	3.05*	2.96*
	(1.66)	(1.61)	(1.62)	(1.57)
Listed	-1.72	-1.19	-0.90	-0.36
	(1.65)	(1.59)	(1.57)	(1.52)
Unlisted	3.06*	3.47**	3.69**	4.12***
	(1.56)	(1.51)	(1.51)	(1.45)
Retail	8.35***	8.69***	8.87***	9.23***
	(1.79)	(1.72)	(1.73)	(1.66)
Contract specific characteristics				
LogTenor	2.88***	2.84***	2.81***	2.77***
	(0.24)	(0.24)	(0.23)	(0.23)
LogNotional	-1.38***	-1.36***	-1.37***	-1.35***
	(0.17)	(0.16)	(0.16)	(0.16)
LogCustomization	1.65***	1.64***	1.62***	1.62***
	(0.23)	(0.22)	(0.22)	(0.22)
Volatility	-5.23	-4.71	-3.01	-2.21
	(4.78)	(4.75)	(4.31)	(4.28)
Sell	7.93***	7.94***	7.58***	7.60***
	(0.55)	(0.54)	(0.55)	(0.54)
FEs				
Dealer FE	Yes	Yes	Yes	Yes
Date FE	No	No	Yes	Yes
Minute-of-day FE	No	Yes	No	Yes
Adjusted R-squared	0.22	0.23	0.24	0.24
Observations	255,036	255,036	255,036	255,036

Table 6: Model Results (Category Wise)

This table describes results for regressions of the markup on measures of client sophistication and contract characteristics, for each client category. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	Retail	Unlisted	Listed
Client specific characteristics			
Log#Dealers	-9.56***	-4.94***	-2.20***
	(1.20)	(0.46)	(0.33)
Contract specific characteristics			
LogTenor	5.60***	2.95***	0.92***
	(0.43)	(0.32)	(0.16)
LogNotional	-2.08***	-1.05***	-0.26**
	(0.32)	(0.16)	(0.11)
LogCustomization	1.97***	1.92***	0.62***
	(0.33)	(0.34)	(0.18)
Volatility	-8.87	-7.98	10.37
	(11.48)	(5.59)	(6.37)
Sell	8.79***	7.39***	3.27***
	(0.97)	(0.75)	(0.52)
FEs			
Dealer FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Minute-of-day FE	Yes	Yes	Yes
Adjusted R-squared	0.37	0.24	0.12
Observations	35,431	125,743	90,865

Table 7: Breakup of Markup paid and Notional

This table provides a breakup of markup and notional by clients dealing one vs. two and more dealers. #Dealers indicate the total number of dealers with which a client has transacted during the observation period. Clients represents the proportion of clients transacting with a specific number of deal counterparties. Notional and markup represent share in gross notional traded and total markup incurred respectively.

#Dealers	Clients (share)	Notional (share)	Markup (share)
1	83%	20%	51%
2 and more	17%	80%	49%

Table 8: Category wise Share in Clients, Notional and Markup

This table provides a breakup of markup and notional by client category. Clients represents the proportion of clients transacting with a specific number of deal counterparties. Notional and markup represent share in gross notional traded and total markup incurred respectively.

Client Category	Clients (share)	Notional (share)	Markup (share)
Unlisted	46%	28%	51%
Retail	44%	3%	14%
Listed	9.1%	57%	25%
Foreign Investor	0.7%	10%	7%
Related Entity	0.1%	2%	2%

Table 9: Model Results (Periods of High and Low Volatility)

This table describes results for regressions of the markup on measures of client sophistication and contract characteristics for periods of high (H) and low (L) volatility. The segregation of observation period into two periods of high and low volatility was done based on the movement of daily rolling three months realised volatility. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	H	L
Client specific characteristics		
Log#Dealers	-3.40***	-4.45***
	(0.45)	(0.49)
Contract specific characteristics		
LogTenor	2.10***	2.23***
	(0.23)	(0.32)
LogNotional	-0.75***	-0.79***
	(0.14)	(0.14)
LogCustomization	1.38***	1.52***
	(0.19)	(0.26)
Volatility	-0.12	-1.51
	(18.92)	(39.12)
Sell	5.45***	8.63***
	(0.49)	(0.62)
Fixed Effects (FEs)		
Dealer FE	Yes	Yes
Date FE	Yes	Yes
Minute-of-day FE	Yes	Yes
Adjusted R-squared	0.20	0.28
Observations	61,450	62,902

Table 10: Model Results (Client Sophistication): Non-Financial Clients

This table describes results for regressions of the markup on measures of client sophistication and contract characteristics for non-financial clients. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	(1)	(2)	(3)	(4)
Client specific characteristics				
Log#Dealers	-4.51***	-4.44***	-4.38***	-4.31***
	(0.52)	(0.51)	(0.50)	(0.49)
Contract specific characteristics				
LogTenor	2.47***	2.46***	2.41***	2.39***
	(0.27)	(0.27)	(0.26)	(0.26)
LogNotional	-0.74***	-0.74***	-0.75***	-0.75***
	(0.16)	(0.15)	(0.15)	(0.15)
LogCustomization	1.70***	1.69***	1.68***	1.67***
	(0.24)	(0.24)	(0.23)	(0.23)
Volatility	-3.66	-3.42	-1.08	-0.55
	(4.81)	(4.79)	(4.23)	(4.22)
Sell	7.68***	7.70***	7.30***	7.31***
	(0.55)	(0.53)	(0.54)	(0.53)
Fixed Effects (FEs)				
Dealer FE	Yes	Yes	Yes	Yes
Date FE	No	No	Yes	Yes
Minute-of-day FE	No	Yes	No	Yes
Adjusted R-squared	0.23	0.23	0.24	0.25
Observations	251,006	251,006	251,006	251,006

Table 11: Model results: Alternative Measures of Client Sophistication

This table describes results for regressions of the markup on three alternative measures of client sophistication. The first one is Log#Transactions which is calculated as the log of the total number of transactions entered into by a client in the sample period. The second measure is LogGrossNotional which is calculated as log of gross notional of all forward contracts traded by a client in the sample period. The third is a Sophistication Index which is calculated as the first principal component of Log#Dealers, Log#Transactions and LogGrossNotional. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	(1)	(2)	(3)
Client specific characteristics			
Log#Transactions	-2.31***		
	(0.19)		
LogGrossNotional		-2.06***	
		(0.17)	
Sophistication Index			-3.23***
			(0.32)
Contract specific characteristics			
LogTenor	2.34***	2.18***	2.20***
	(0.26)	(0.26)	(0.27)
LogNotional	-1.41***	-0.15	-0.70***
	(0.18)	(0.12)	(0.15)
LogCustomization	1.69***	1.57***	1.61***
	(0.21)	(0.20)	(0.21)
Volatility	-2.89	-3.23	-2.46
	(4.27)	(4.27)	(4.27)
Sell	7.35***	6.90***	7.09***
	(0.48)	(0.47)	(0.48)
Fixed Effects (FEs)			
Dealer FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Minute-of-day FE	Yes	Yes	Yes
Adjusted R-squared	0.25	0.27	0.26
Observations	255,352	255,352	255,352

Table 12: Model Results: Relationship Trading

This table describes results for regressions of the markup on relationship trading of client and dealer pairs. Markup is defined as the difference between forward contract rate and contemporaneous interbank outright forward mid-price. Relationship dummy takes a value of one if the dealer was listed as a banker for a given client and zero, otherwise. #Dealers is defined as the number of dealer counterparties with which a client has traded in USD/INR forwards during sample period. LogTenor is the natural logarithm of number of days of tenor of the forward contract. LogNotional is the natural logarithm of the notional value of the forward contract. LogCustomization is the natural logarithm of the difference in days of the settlement date of the forward contract, from the nearest month end. Volatility is calculated using Rogers-Satchell Volatility estimator during each half hour interval. Sell is a dummy variable which takes a value 1 and 0 depending upon whether the client is short or long USD respectively. *, **, *** denote statistical significance at the 10 percent, 5 percent, and 1 percent levels respectively. Standard errors clustered at the client level are reported in parentheses.

	(1)	(2)	(3)
Client specific characteristics			
Relationship	-0.62	-1.31**	-1.08
	(0.65)	(0.65)	(1.83)
Log#Dealers		-2.80***	-2.69***
		(0.56)	(0.81)
Relationship * Log#Dealers			-0.13
			(0.76)
Contract specific characteristics			
LogTenor	1.57***	1.14***	1.14***
	(0.21)	(0.26)	(0.26)
LogNotional	-0.73***	-0.28*	-0.29**
	(0.14)	(0.15)	(0.14)
LogCustomization	0.96***	0.85***	0.85***
	(0.23)	(0.21)	(0.21)
Volatility	3.91	4.95	4.96
	(6.26)	(6.23)	(6.23)
Sell	4.24***	3.73***	3.73***
	(0.72)	(0.71)	(0.71)
Fixed Effects (FEs)			
Dealer FE	Yes	Yes	Yes
Date FE	Yes	Yes	Yes
Minute-of-day FE	Yes	Yes	Yes
Adjusted R-squared	0.15	0.17	0.17
Observations	82,412	82,412	82,412